

available, too. There is also a radio driver to which you can connect through a transceiver module, which acts as a multiplexer between the network stack and the driver.

Compatible with a number of hardware chains

RIOT offers support to a whole lot of microcontrollers like MSP430, ARM7, Cortex series and x86 processors and a number of boards like Arduino boards, mbed NXP LPC1768, Nordic chip, Spark-Core, STM family and Texas Instruments development kit, to pick a few from the long list.

You only need to install a tool-chain for the corresponding hardware before you can begin working

RIOT provides cryptographic libraries and data structures like bloom filters, hash tables and priority queues. A shell acts as the link between the brain of the developer and that of the OS. To make sure that your code is of high quality, you can employ established tools like Travis and embUnit-Unit Testing.

Run RIOT from your computer

RIOT provides a native port that allows you to run the OS as a process on a Linux or MAC OS. For those who are not too familiar with embedded programming, or those who do not have one of the supported platforms, this turns out to be a handy feature.

Getting started

It is easy to start working with RIOT. Falling under the free software category, you can simply download the package and begin coding right away. The package provides example applications to help you get acquainted with the software and even a sample Makefile template that shows you how to compile your code.

Wide community support, active presence on twitter (@RIOT_OS), videos on YouTube and Vimeo and, not to forget, an informative website (<http://riot-os.org/>) provide the needed support as you begin your venture operating this OS.

A good place to start would be <https://github.com/RIOT-OS/RIOT/wiki/Creating-your-first-RIOT-project>

with the setup. Make sure you also define the target hardware in your Makefile. Just like you call modules in Verilog coding, you can include modules for whatever interfaces you want to have in your design and add this to the Makefile as well. Unless you meddle with the settings, the designers have developed the software to automatically create an elf file and an Intel hex file as results of your application, which come in handy.

Utilities for a comfortable design experience

The most important factor while coding is the amount of resources available. Even though you might be coding in a familiar language, if the constructs are not recognised or supported by the tool being used, your expertise comes to naught.

With this port, you can develop and debug projects written for RIOT and the OS itself, without the limitations and requirements of debugging on actual hardware. Use tools like gdb or valgrind to analyse your process. You can even exploit a configurable topology to create virtual test beds of multiple networked instances of RIOT running simultaneously, or create a controlled environment and experiment your system's functioning there.

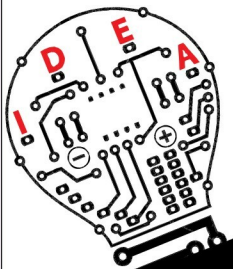
Apart from these, RIOT is also compatible with Docker; you have everything in one place and setting up the development environment is easy. The only thing left now is to try your hand at this tool. This month's DVD has the package you need—begin exploring right away. **EFY**

Priya Ravindran is a freelance writer with EFY

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